Spatial Filtering

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Domain of image processing

- ❑ Spatial Domain (Direct operation on image)
- ❑ Frequency domain (Operation after Fourier transformation)

Spatial processing

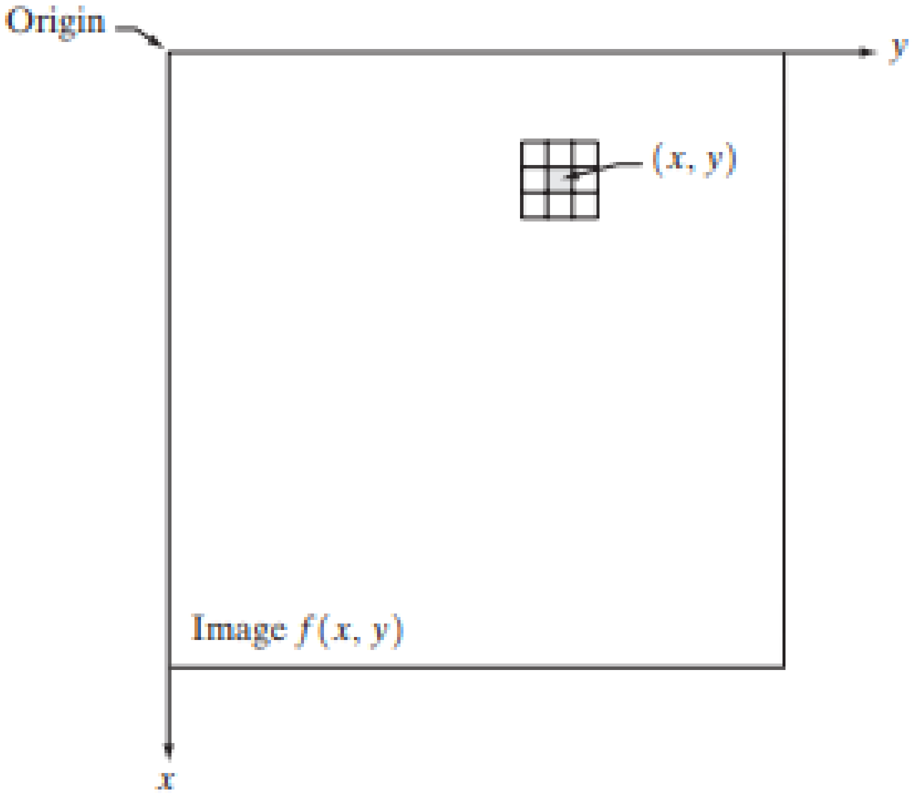
$$g(x,y) = \mathrm{T}\,[\,f(x,y)]$$

- ❑ $f(x,y)$ Input Image
- ❑ $g(x,y)$ Output image
- ❑ T Operation/ Transform method

Spatial processing

$$g(x,y) = T\,[\,f(x,y)]$$

- ❑ *f(x,y)* Input Image
- ❑ *g(x,y)* Output image
- ❑ T Operation/ Transform method defined over the neighbourhood of point(x,y)



Spatial processing

For smallest neighbourhood of size 1 * 1, g depends only on value of f at a single point (x,y) and T becomes a intensity transformer function. Also called gray level mapping.

Hence for simplicity of notation it is described by

$$s = T\ (r)$$

Here s and r denotes intensity of g and f at a point (x,y)

Spatial processing

- Intensity transformation

  (Operates on single pixel)

- Spatial Filtering

  (Operates in neighbourhood)

Spatial filtering

A spatial filter consists of

(a) **a neighborhood**, and

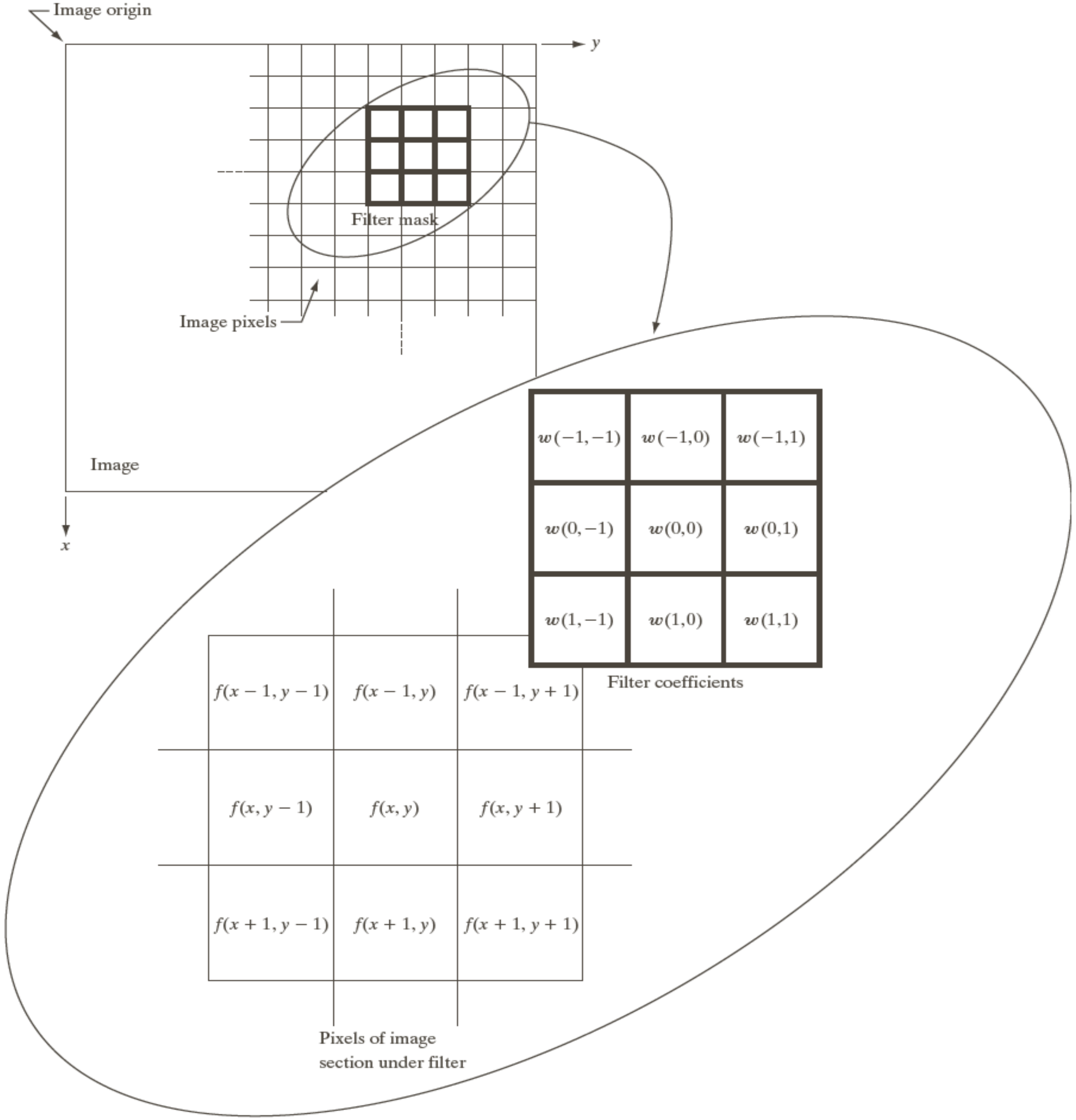
(b) **a predefined operation**

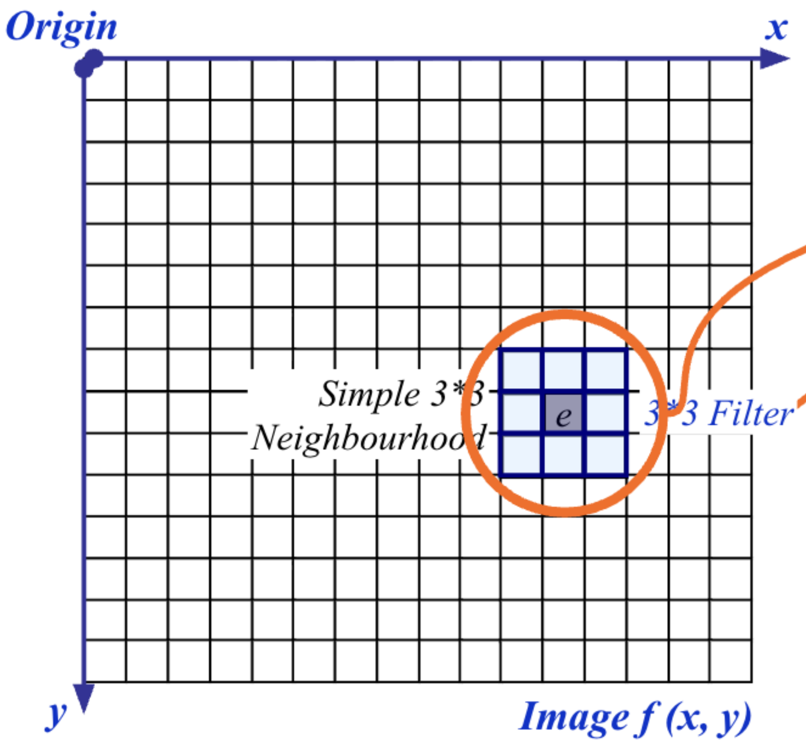
Linear spatial filtering of an image of size **M*N** with a filter of size **m*n** is given by the expression-

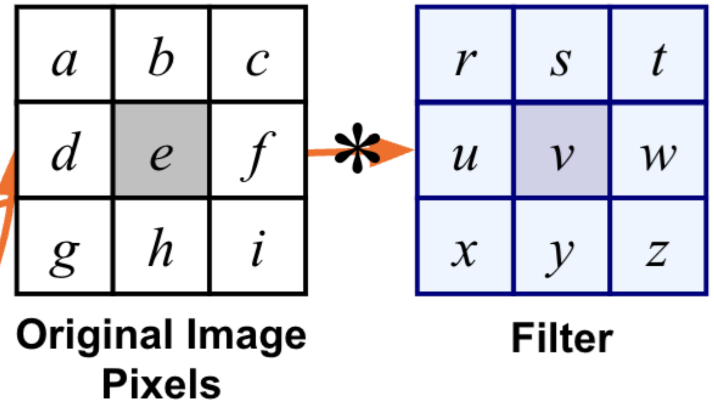
$$g(x,y)=\sum_{s=-a}^{a}\sum_{t=-b}^{b} w(s,t)f(x+s,y+t)$$

Here, a = (m-1)/2 and b=(n-1)/2

Spatial filtering







$$
\begin{aligned}
e_{processed} = {} & v*e + \\
& r*a + s*b + t*c + \\
& u*d + w*f + \\
& x*g + y*h + z*i
\end{aligned}
$$

The above is repeated for every pixel in the original image to generate the filtered image

Spatial filtering

Correlation

The correlation of a filter $w(x, y)$ of size $m \times n$ with an image $f(x, y)$, denoted as $w(x, y) \quad f(x, y)$

$$w(x, y) \star f(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) f(x + s, y + t)$$

Spatial filtering

Convolution

The convolution of a filter $w(x, y)$ of size $m \times n$ with an image $f(x, y)$, denoted as $w(x, y) \quad f(x, y)$

$$w(x, y) \bigstar f(x, y) = \sum_{s=-a}^{a} \sum_{t=-b}^{b} w(s, t) f(x - s, y - t)$$

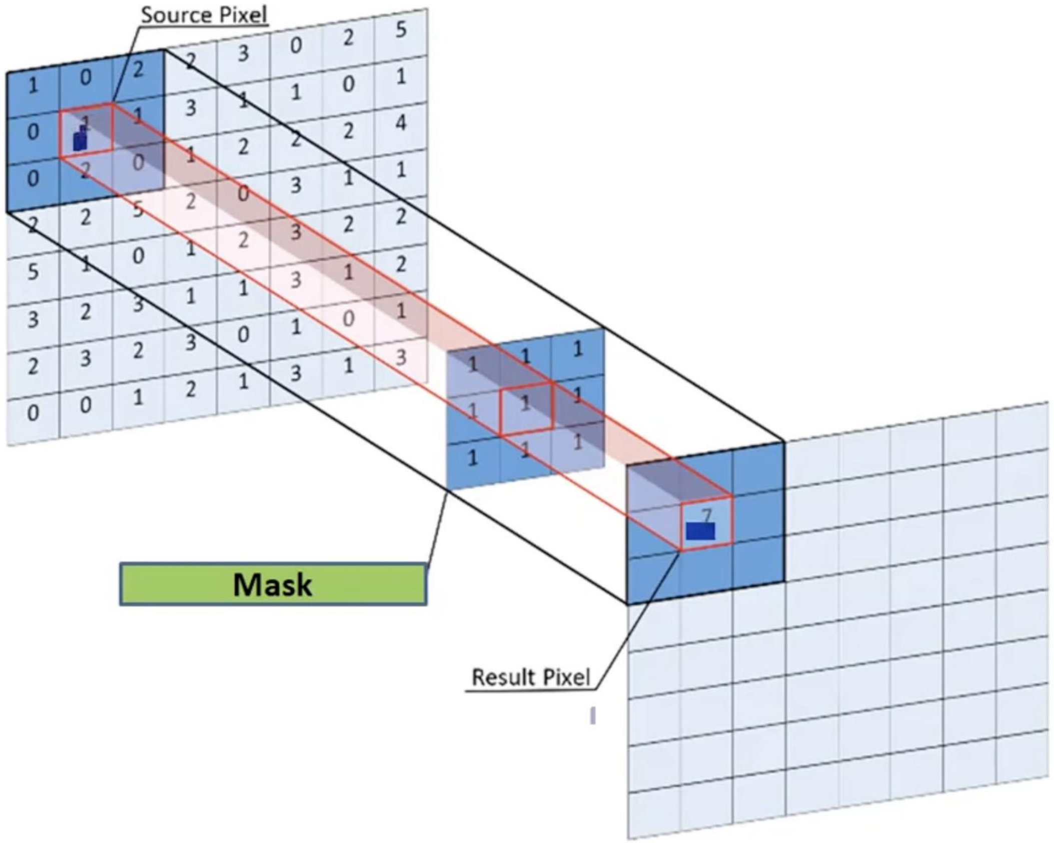
Spatial filtering

Convolution

In order to perform convolution on an image, following steps should be taken.

- Flip the mask (horizontally and vertically) only once
- Slide the mask onto the image.
- Multiply the corresponding elements and then add them
- Repeat this procedure until all values of the image has been calculated.

Performing Convolution



Spatial filtering

Convolution

25	100	75	49	130
50	80	0	70	100
5	10	20	30	0
60	50	12	24	32
37	53	55	21	90
140	17	0	23	222

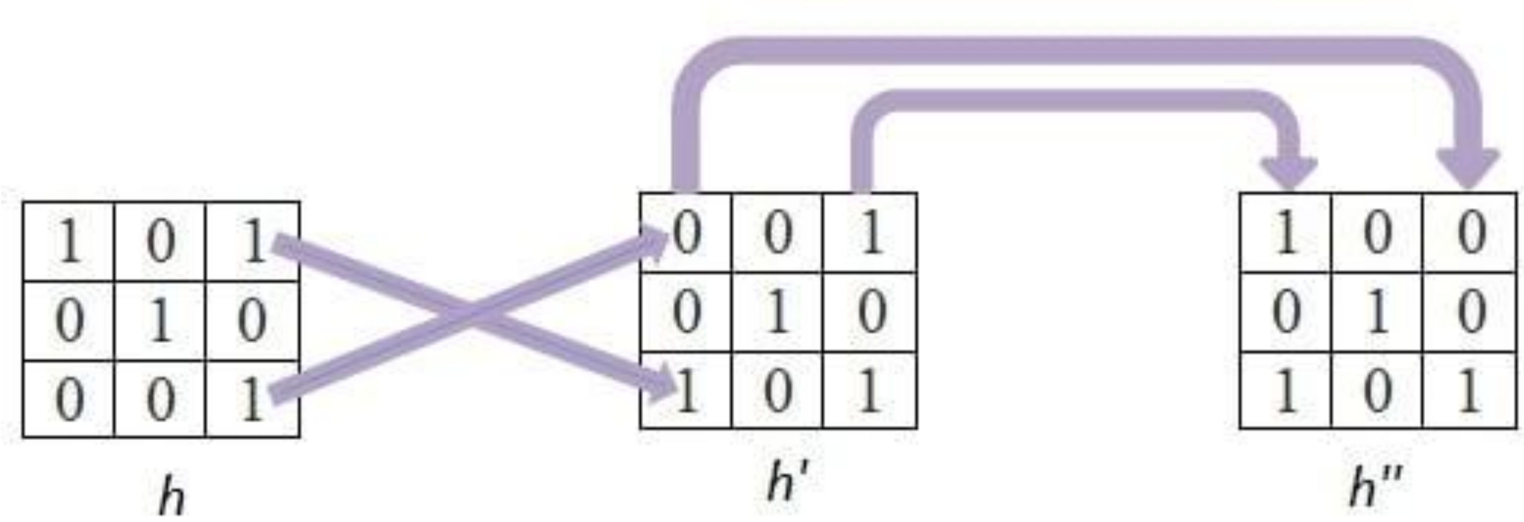
x

1	0	1
0	1	0
0	0	1

h

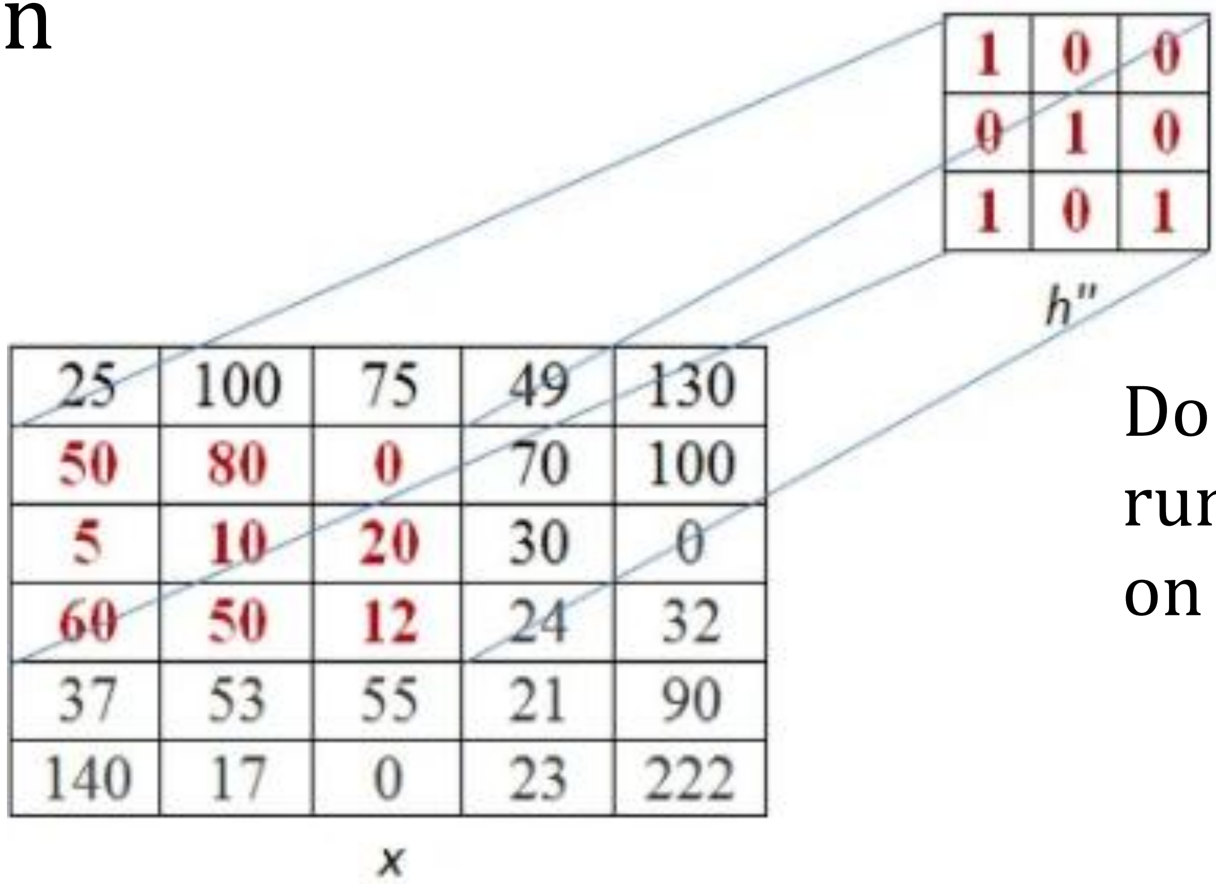
Spatial filtering

Convolution



Spatial filtering

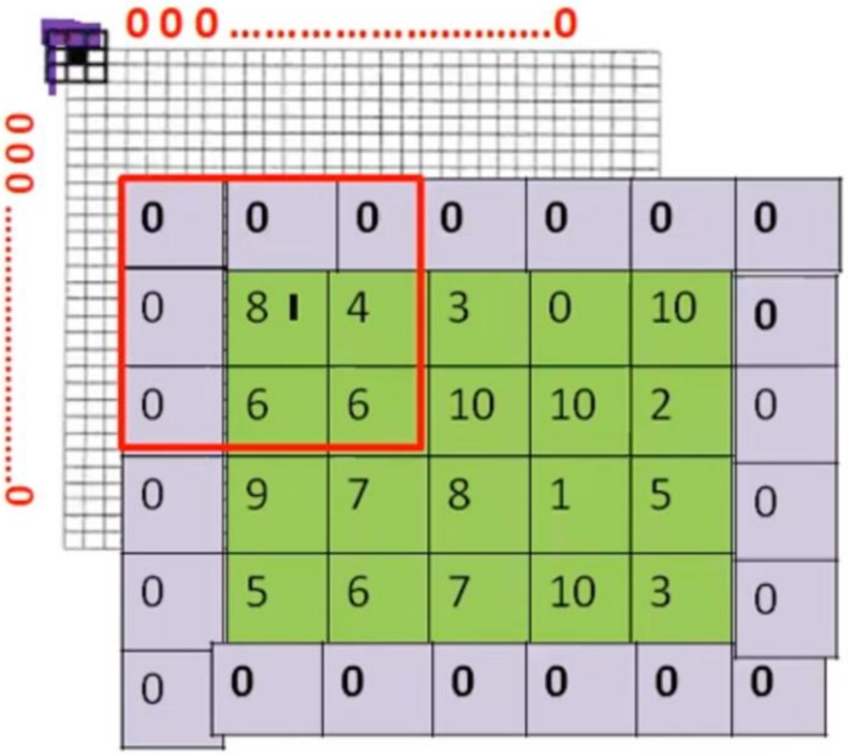
Convolution



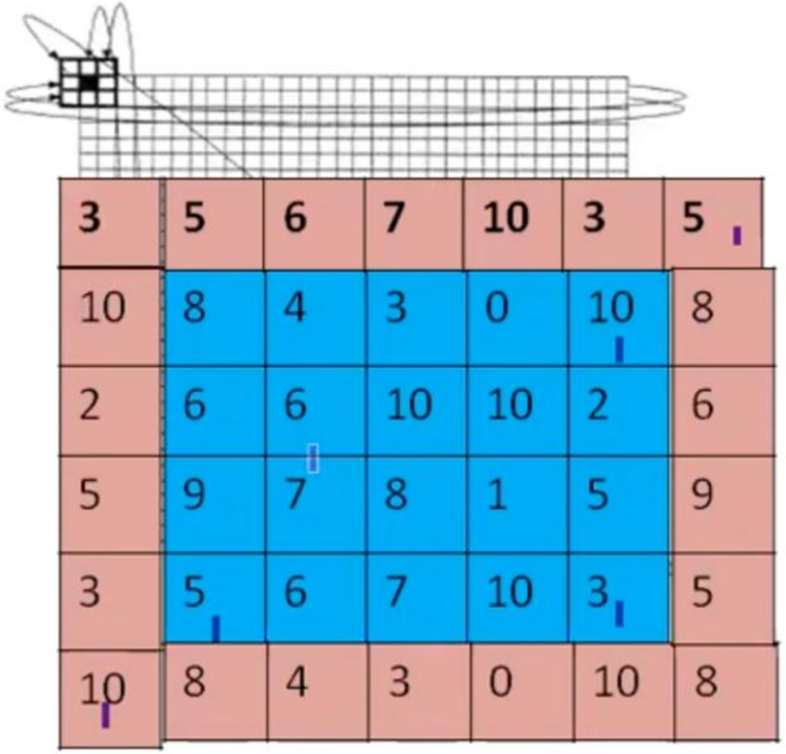
Do padding for running the filter on boundary pixels

Handling Boundary Pixels

pad with zeroes



wrap around



Spatial filtering

MASK

A filter is also known as a mask. Concept of spatial filtering is also known as masking. It is done by convolving a mask with an image.

Filters are applied on image for multiple purposes. The two most common uses are as following:

- Filters are used for Blurring and noise reduction
- Filters are used or edge detection and sharpness

Smoothing Spatial Filters (Linear)

One of the simplest spatial filtering operations we can perform is a smoothing operation

- Simply average all of the pixels in a neighbourhood around a central value
- Especially useful in removing noise from images
- Also useful for highlighting gross detail

$^1/_9$	$^1/_9$	$^1/_9$
$^1/_9$	$^1/_9$	$^1/_9$
$^1/_9$	$^1/_9$	$^1/_9$

Simple averaging filter

Smoothing Spatial Filters

Applications of Smoothing Filters:

Smoothing filters are used in image processing for the following purposes:

1. **Noise Reduction:**
   Removes unwanted noise such as salt-and-pepper noise from images

2. **Blurring:**
   Softens the image by reducing sharp intensity variations

3. **Removal of Small Details:**
   Eliminates fine textures and minor variations

4. **Bridging Gaps:**
   Fills small breaks in lines and curves

5. **Preprocessing:**
   Used before edge detection and segmentation to improve results

Spatial filtering

Smoothing Spatial Filters

$$\frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

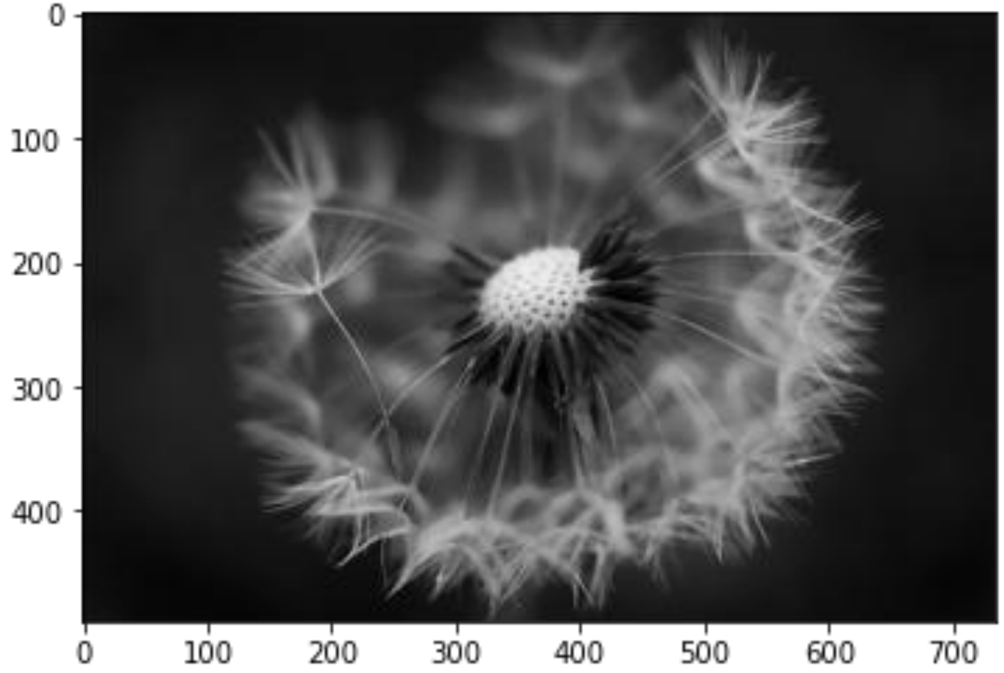
Box filter

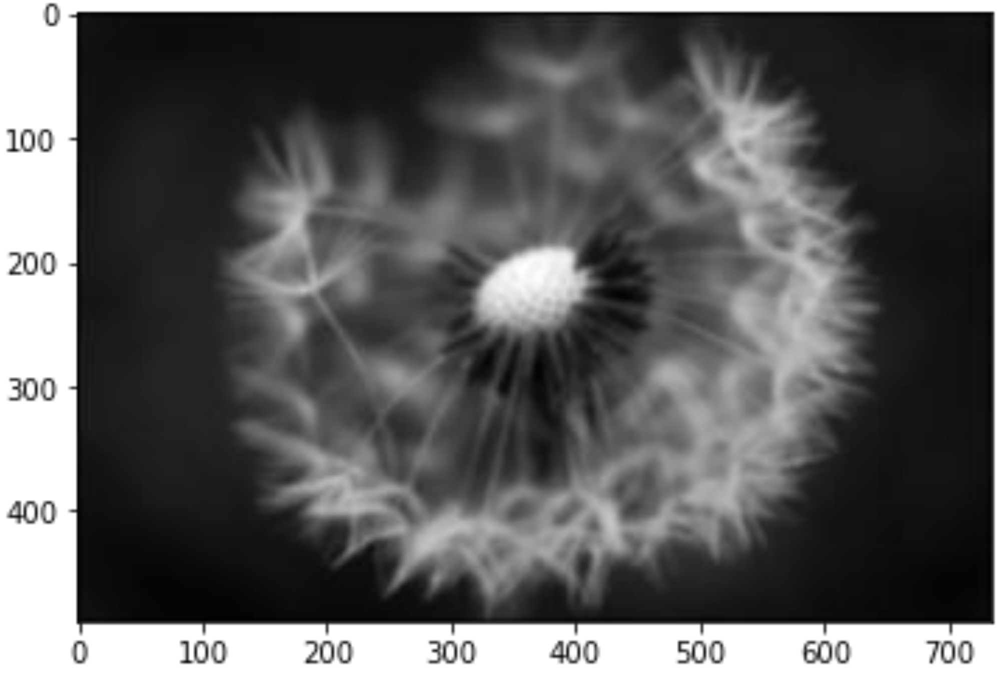
$$\frac{1}{16} \times \begin{bmatrix} 1 & 2 & 1 \\ 2 & 4 & 2 \\ 1 & 2 & 1 \end{bmatrix}$$

Weighted average filter

Spatial filtering

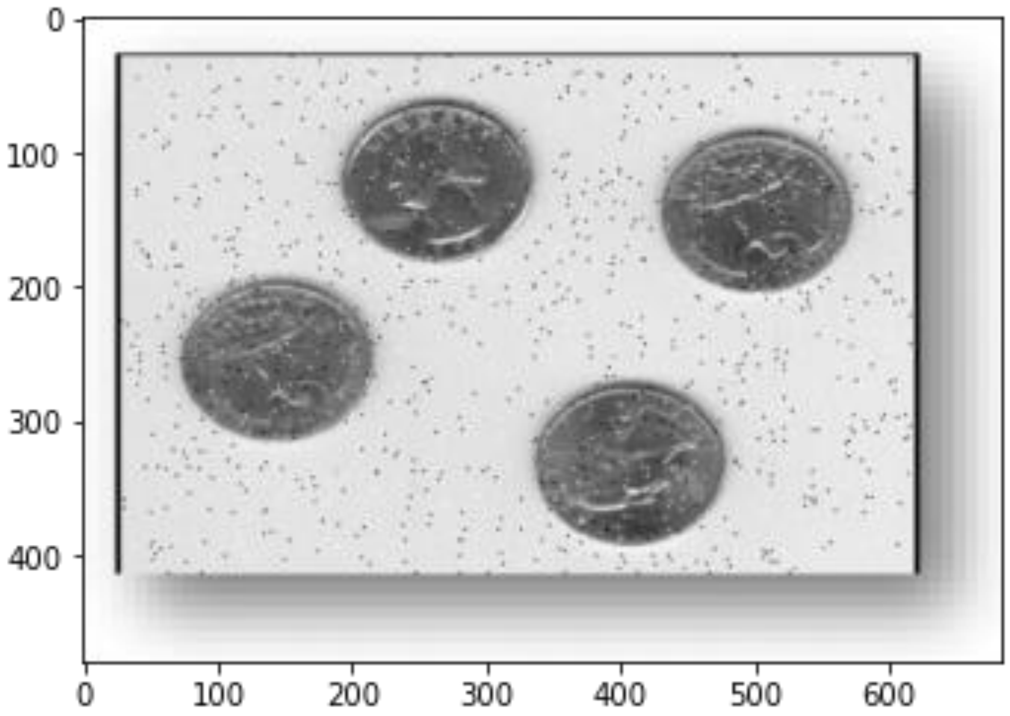
Smoothing Spatial Filters (Blurring)

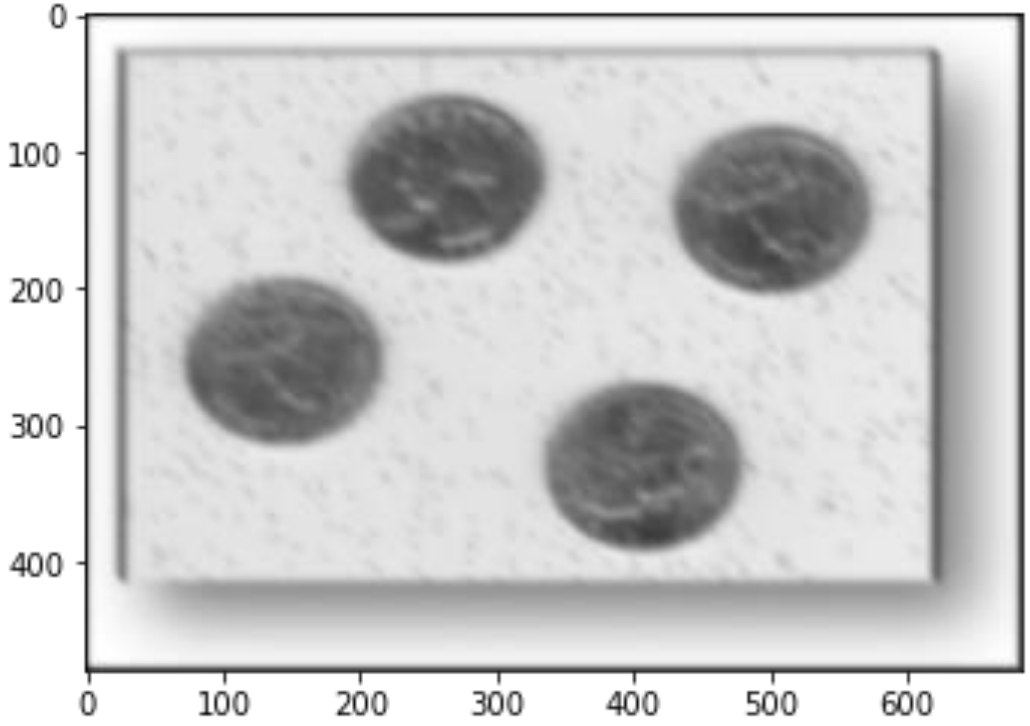




Spatial filtering

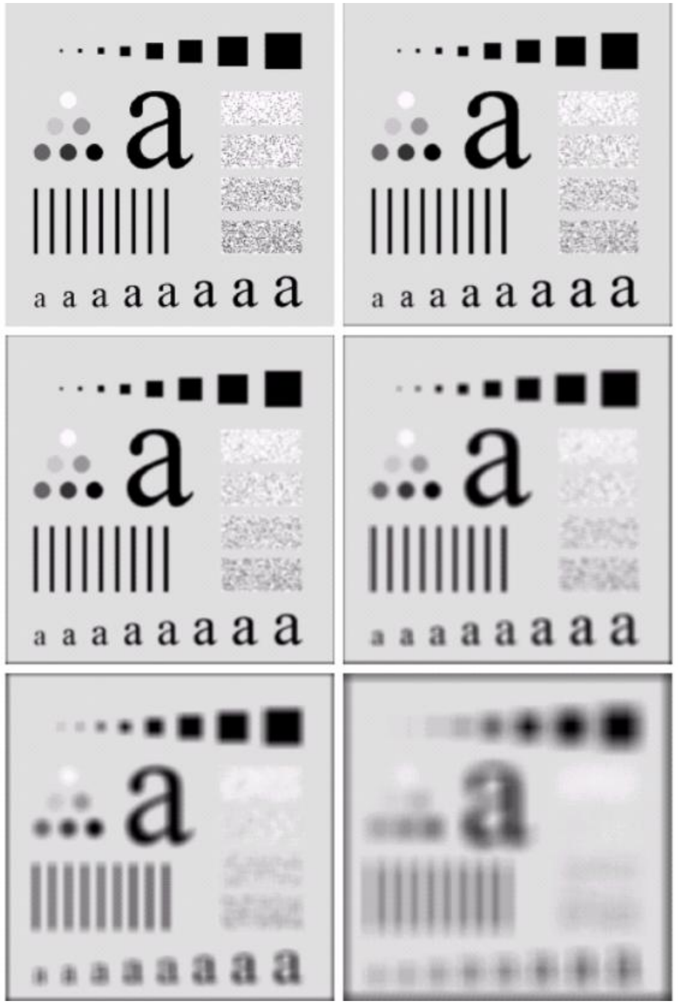
Smoothing Spatial Filters (Noise reduction)





Effect of Filter size

- The image at the top left is an original image of size 500*500 pixels
- The subsequent images show the image after filtering with an averaging filter of increasing sizes
  – 3, 5, 9, 15 and 35
- Notice how detail begins to disappear



Weighted Smoothing Filter

More effective smoothing filters can be generated by allowing different pixels in the neighbourhood different weights in the averaging function

- Pixels closer to the central pixel are more important
- Often referred to as a *weighted averaging*

$^{1}/_{16}$	$^{2}/_{16}$	$^{1}/_{16}$
$^{2}/_{16}$	$^{4}/_{16}$	$^{2}/_{16}$
$^{1}/_{16}$	$^{2}/_{16}$	$^{1}/_{16}$

Weighted averaging filter

Spatial filtering

Order-statistic (Nonlinear) Filters

— Nonlinear

— Based on ordering (ranking) the pixels contained in the filter mask

— Replacing the value of the center pixel with the value determined by the ranking result

E.g., median filter, max filter, min filter

1. Consider the image below and calculate the output of the pixel (2,2) if smoothing is done using 3 x 3 neighbourhood using all the filters below:

a) Box / Mean filter
b) Weighted average filter
c) Median filter
d) Min filter
e) Max filter

1	8	8	0	7
4	7	9	5	7
5	4	6	8	6
4	2	0	1	5
0	1	0	2	0

3x3

a) Box filter

$$= \frac{1}{9} \times [7+9+5+4+6+8+2+0+1] \qquad \frac{1}{9} \times \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}$$

$$= \frac{1}{9} \times [42] = 4.66 \approx \underline{5}.$$

b) Weighted average filter

$$= \frac{1}{16}\left[7\times1 + 9\times2 + 5\times1 + 4\times2 + 4\times6 + 8\times2 + 2\times1 + 0\times2 + 1\times1\right]$$

$$\frac{1}{16}\times\begin{bmatrix}1 & 2 & 1\\ 2 & 4 & 2\\ 1 & 2 & 1\end{bmatrix}$$

$$= \frac{1}{16}\left[81\right] = 5.0625 \approx \underline{\underline{5}}.$$

c) Median filter

$\cancel{0}, \cancel{1}, \cancel{2}, \cancel{4}, 5, \cancel{6}, \cancel{7}, \cancel{8}, \cancel{9}$

Median = $\underline{\underline{5}}$.

d) Min filter

= $\underline{\underline{0}}$.

e) Max filter

= $\underline{\underline{9}}$.

Q. Why Median filter is better than mean filter?

Median filter is normally used to reduce noise in an image, similar to the mean filter. However, it often does a better job than mean filter in preserving useful detail in an image.

Median filter has 2 main advantages:

1) The median is a more robust average than the mean and so a single very unrepresentative pixel in a neighborhood will not affect the median value significantly.

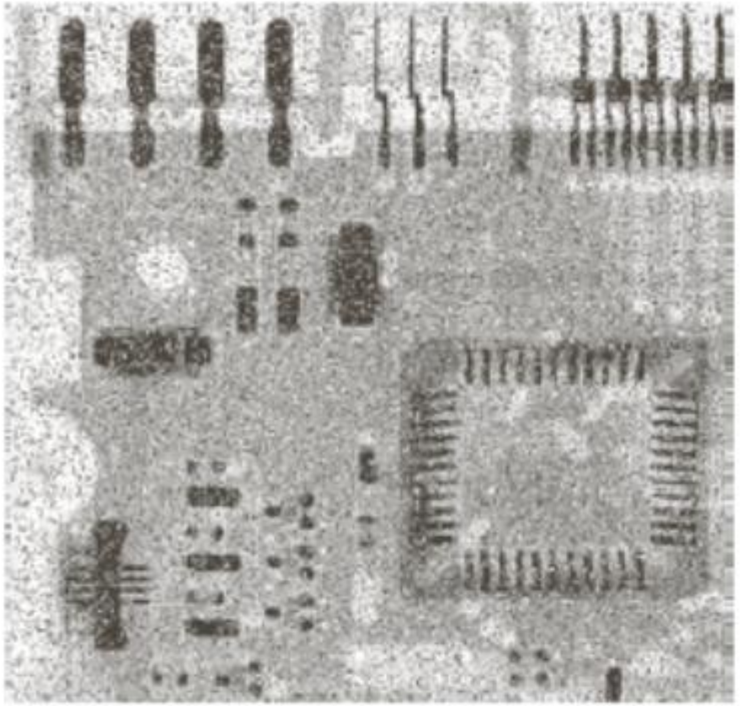
2) Since the median value must actually be the value of one of the pixels in the neighborhood, the median filter does not create new unrealistic pixel values when the filter straddles an edge. Therefore, it is much better at preserving sharp edges than the mean filter.

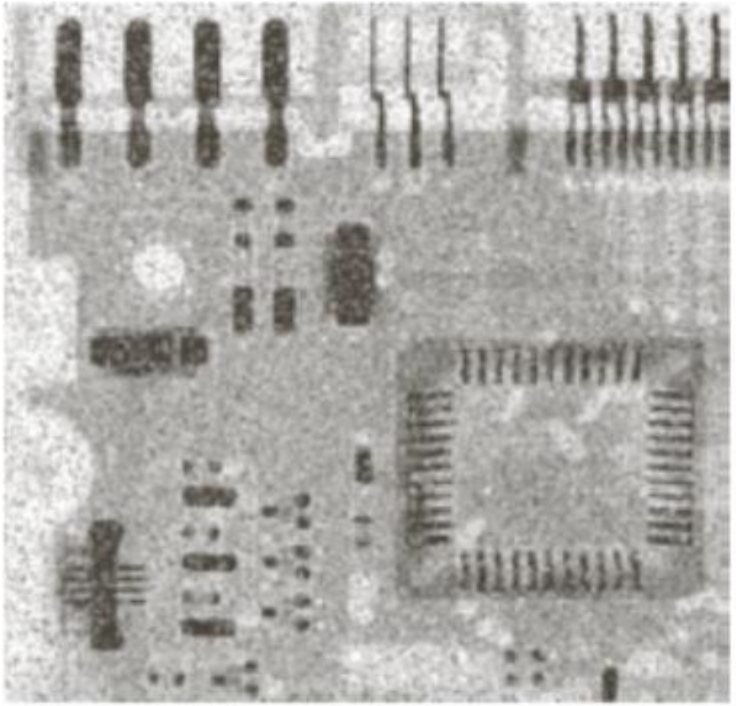
Note:

- Mean filter is better at dealing with Gaussian noise than median filter.
- Median filter is better at dealing with salt and pepper noise than mean filter.

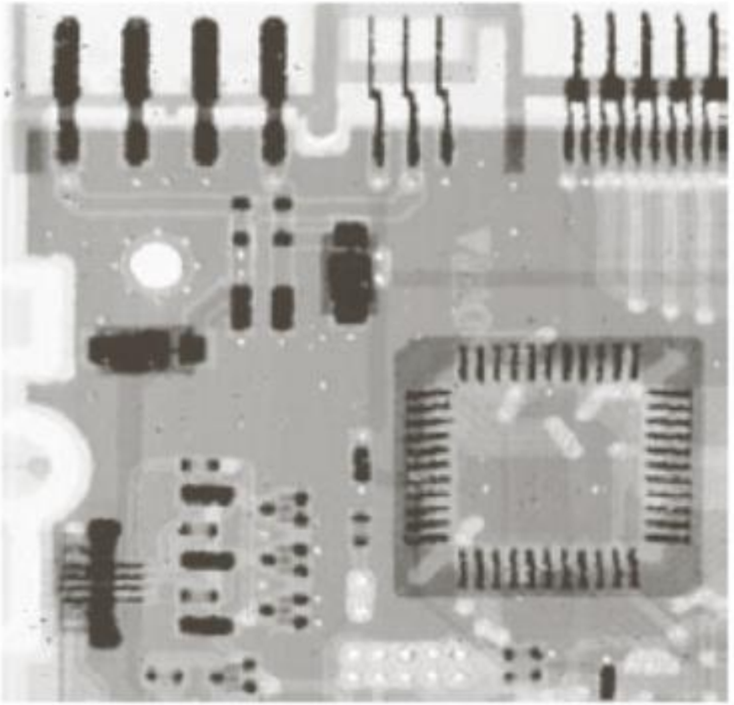
Spatial filtering

Order-statistic (Nonlinear) Filters





Mean filter



Median filter

Sharpening Spatial Filters

- Foundation
- Laplacian Operator
- Unsharp Masking and Highboost Filtering
- Using First-Order Derivatives for Nonlinear Image Sharpening
  - The Gradient

Sharpening Spatial Filters

Foundation

- The first-order derivative of a one-dimensional function f(x) is the difference

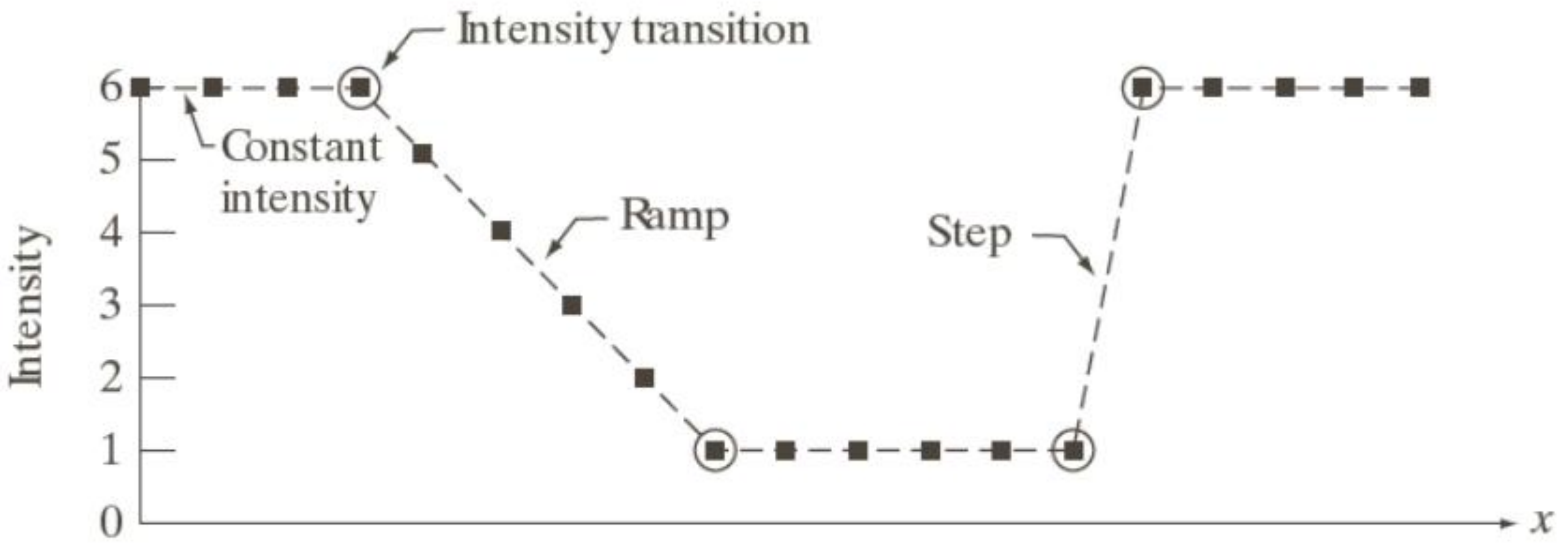
$$\frac{\partial f}{\partial x} = f(x+1) - f(x)$$

- The second-order derivative of f(x) as the difference

$$\frac{\partial^2 f}{\partial x^2} = f(x+1) + f(x-1) - 2f(x)$$

Sharpening Spatial Filters

Foundation



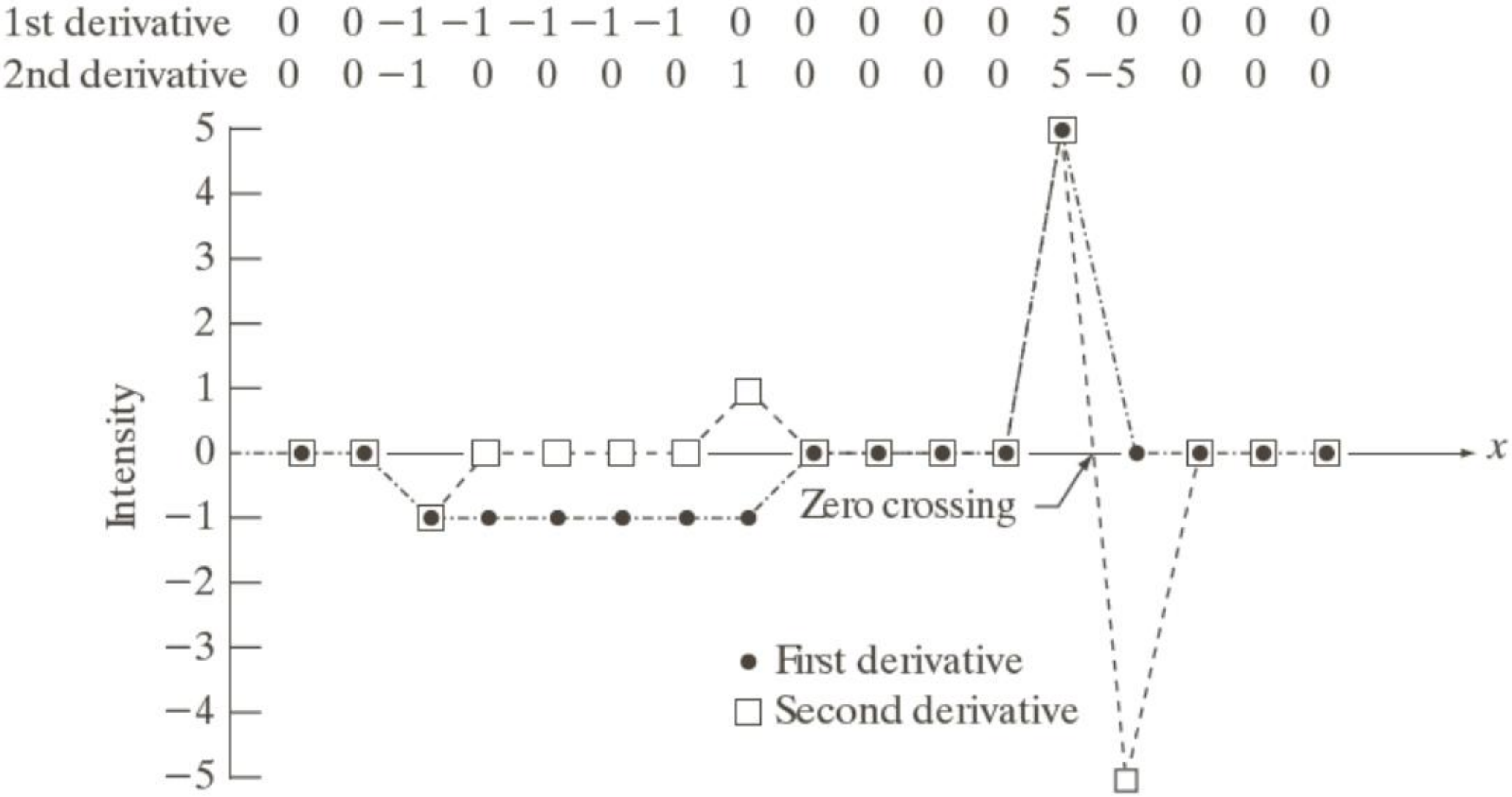
	Scan line	6	6	6	6	5	4	3	2	1	1	1	1	1	1	6	6	6	6	6	→ x
$f(x+1) - f(x)$	1st derivative		0	0	−1	−1	−1	−1	−1	0	0	0	0	0	5	0	0	0	0		
$f(x+1) + f(x-1) - 2f(x)$	2nd derivative		0	0	−1	0	0	0	0	1	0	0	0	0	5	−5	0	0	0		

Sharpening Spatial Filters

Foundation

$$f(x+1) - f(x)$$

$$f(x+1) + f(x-1) - 2f(x)$$



Sharpening Spatial Filters

Foundation

►First-order derivatives generally produce thicker edges in image

►Second-order derivatives have a stronger response to fine detail, such as thin lines, isolated points, and noise

►Second-order derivatives produce a double-edge response at ramp and step transition in intensity

►The sign of the second derivative can be used to determine whether a transition into an edge is from light to dark or dark to light

Sharpening Spatial Filters

The second-order isotropic derivative operator is the **Laplacian** for a function (image) f(x,y)

$$\nabla^2 f = \frac{\partial^2 f}{\partial x^2} + \frac{\partial^2 f}{\partial y^2}$$

$$\frac{\partial^2 f}{\partial x^2} = f(x+1, y) + f(x-1, y) - 2f(x, y)$$

$$\frac{\partial^2 f}{\partial y^2} = f(x, y+1) + f(x, y-1) - 2f(x, y)$$

$$\nabla^2 f = f(x+1, y) + f(x-1, y) + f(x, y+1) + f(x, y-1) - 4f(x, y)$$

Sharpening Spatial Filters

Laplacian mask

0	1	0
1	−4	1
0	1	0

1	1	1
1	−8	1
1	1	1

0	−1	0
−1	4	−1
0	−1	0

−1	−1	−1
−1	8	−1
−1	−1	−1

Sharpening Spatial Filters

Image sharpening in the way of using the Laplacian:

$$g(x,y) = f(x,y) + c[\nabla^2 f(x,y)]$$

where,

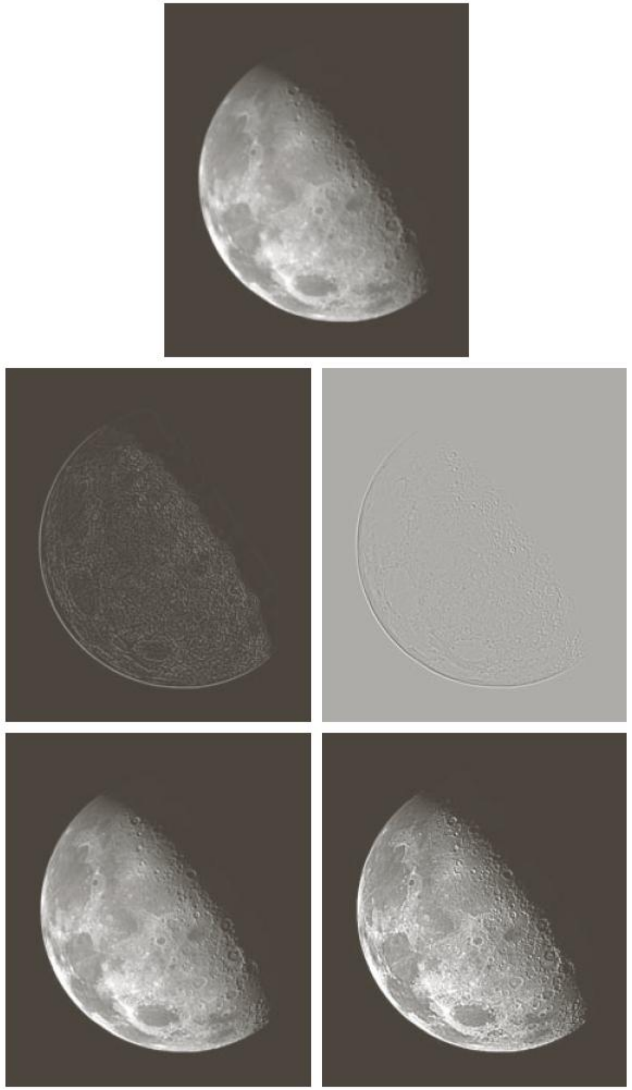
$f(x,y)$ is input image,

$g(x,y)$ is sharpenend images,

$c = -1$ if $\nabla^2 f(x,y)$ corresponding to 2 $positive\ mask$

Sharpening Spatial Filters

Image sharpening in the way of using the Laplacian



Sharpening Spatial Filters

Unsharp Masking and Highboost Filtering

- Unsharp masking

  Sharpen images consists of subtracting an unsharp (smoothed) version of an image from the original image

- Steps

1. Blur the original image
2. Subtract the blurred image from the original
3. Add the mask to the original

Sharpening Spatial Filters

Unsharp Masking and Highboost Filtering

Let $\overline{f}(x, y)$ denote the blurred image, unsharp masking is

$$g_{mask}(x, y) = f(x, y) - \overline{f}(x, y)$$

Then add a weighted portion of the mask back to the original

$$g(x, y) = f(x, y) + k * g_{mask}(x, y) \qquad k \geq 0$$

when $k > 1$, the process is referred to as highboost filtering.

Sharpening Spatial Filters

Unsharp Masking and Highboost Filtering